



Study on phase transition and contrast-enhanced imaging of ultrasound-responsive nanodroplets with polymer shells

Yaqiong Li^a, Ruiqing Liu^b, Luwen Liu^c, Ye Zhang^c, Jun Sun^a, Peizhi Ma^a, Yuejin Wu^c, Shaobo Duan^c, Lianzhong Zhang^{c,*}

^a Department of Pharmacy, Henan Provincial People's Hospital, People's Hospital of Zhengzhou University, School of Clinical Medicine, Henan University, Zhengzhou, Henan, 450003, China

^b Department of Interventional Therapy, Henan Provincial People's Hospital, People's Hospital of Zhengzhou University, School of Clinical Medicine, Henan University, Zhengzhou, Henan, 450003, China

^c Department of Ultrasound, Henan Provincial People's Hospital, People's Hospital of Zhengzhou University, School of Clinical Medicine, Henan University, Zhengzhou, Henan, 450003, China

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ABSTRACT

Ultrasound-responsive nanodroplets show great potential in ultrasound diagnosis and targeted tumour therapy due to their unique phase transition properties. However, the mechanism underlying the phase transition and the properties of contrast-enhanced imaging have not been well elucidated, which impedes the development and application of nanodroplets in clinic. Herein, the phase transition of polymeric nanodroplets with a core of perfluoropentane (PFP) was studied through the measurement of particle size and *in vitro/in vivo* contrast-enhanced imaging, and imaging performance was further evaluated by introducing intensity analysis of acoustic signals. The average particle size of nanodroplets increased and became polydispersed when heated at 37 °C, which may result from vaporization of a portion of nanodroplets. For imaging *in vitro*, no acoustic signals were observed at 25 °C when the mechanical index (MI) varied from 0.08 to 1.0. At 37 °C, acoustic signals were observed for MI ≥ 0.4, and the intensity was stronger for higher MIs. For imaging in mice livers, the nanodroplets showed similar contrast enhancement behaviours with SonoVue® at low MI (0.08), which produced strong acoustic signals immediately and were cleared within 10 min. The acoustic signals at high MI (1.0) were weaker but lasted more than 1 h. These results indicated that the phase transition of polymeric nanodroplets could be induced by diagnostic ultrasound, and contrast-enhanced imaging is closely related to particle size, temperature and MI. This study provides a better understanding of phase transition and contrast-enhanced imaging for ultrasound-responsive nanodroplets with polymer shells.

1. Introduction

Microbubbles have been widely used as contrast agents for ultrasound in clinic, and the biophysical effects caused by the cavitation of microbubbles have been extensively explored. The stable and inertial cavitation of microbubbles can trigger mechanical effects such as pore formation, microstreaming, ion channel activation and membrane deformation, which enhance the permeability of the vascular wall and cell membrane and facilitate drug diffusion into the cell [1–4]. Therefore, ultrasound-triggered drug delivery has become a hot spot in the research of tumour targeting therapy [5–8]. However, the shortcomings of gas-filled microbubbles, such as short circulation time and large particle size, not only hinder their effective permeation into the tumour, but also limit their capacity for loading therapeutic agents. To

address these problems, nanodroplets with phase transition properties have been designed to take advantage of the well-studied enhanced permeability and retention (EPR) effect of solid tumours [9–11]. Through the EPR effect, nanodroplets may diffuse out of the vasculature and accumulate within tumour tissues by passive targeting before undergoing phase transition. When exposed to ultrasound, the liquid core of phase-changing nanodroplets vaporizes to the gas phase upon activation by the ultrasound energy, a phenomenon called acoustic droplet vaporization (ADV) [12]. This would effectively extend microbubble-based imaging beyond the vascular space and create new opportunities for tumour detection and treatment [13–16].

Nanodroplets composed of polymer shells and perfluorocarbon (PFC) cores have attracted a great amount of attention in tumour targeting therapy due to their stability and flexible drug loading mode.

* Corresponding author at: No. 7, Weiwu Rd., Jinshui District, Zhengzhou, Henan, 450003, China.

E-mail address: llw0624@gs.zzu.edu.cn (L. Zhang).

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Compounds found within the PFC family differ in configuration and chain length, giving rise to unique molecular weights, densities and boiling points. PFP, in particular, has been the most often used liquid core for droplets. Because its boiling point of 29 °C allows for the possibility of generation of the droplets at room temperature, and the droplets have a lower vaporization threshold at physiological temperature. Nanodroplets based on PFP have been studied for targeted and enhanced release of DOX in tumour tissue with focused ultrasound [17,18].

For ultrasound-sensitive nanodroplets designed for drug delivery, *in vivo* contrast-enhanced imaging performance relying on the phase transition plays an important role in diagnosis and targeted drug release. The distribution and concentration of drug-loaded droplets *in vivo* could be tracked through ultrasound imaging, and the targeting effect could be evaluated. The threshold of vaporization and cavitation is closely related to the imaging performance and side effects of the therapy. The vaporization of droplets at the micron level has been widely studied through *in vitro* experiments and mathematical models [12,19]. However, research on nanodroplets, especially *in vivo*, is still limited. The mechanism underlying the phase transition of nanodroplets has not been well explained. An in-depth study of this aspect would be of great significance for the design and clinical application of novel nanodroplets.

In this study, nanodroplets were constructed based on the degradable amphiphilic copolymer PEG-PCL and PFP. The phase transition behaviour of the nanodroplets was studied through *in vitro* and *in vivo* imaging experiments, and the intensity of ultrasound signals was analysed. Furthermore, the mechanism of the phase transition was extensively discussed. This study could provide guidance for the future design and application of nanodroplets.

2. Material and methods

2.1. Material

Copolymer poly(ethyleneoxide)-block-poly(caprolactone) (PEG-PCL) was purchased from Ponsure Biotech Inc. (Shanghai, China). PEG-PCL had a total molecular weight of 4600 Da; the molecular weights of a PEG block and a PCL block were 2000 Da and 2600 Da, respectively. PFP was obtained from Strem Chemicals, Inc. (Newburyport, USA). Ultrasound contrast agent SonoVue® was purchased from Bracco Suisse SA (Plan-Les-Ouates, Switzerland). All other reagents were of analytical grade.

2.2. Preparation of nanodroplets

First, copolymer micelles were formed using a solvent exchange method [18]. The PEG-PCL copolymer was dissolved in dimethylsulfoxide (DMF), and the solution was added dropwise into PBS under stirring. The mixed solution was dialyzed through a membrane with a molecular weight cutoff of 3000 Da (MYM Biological Technology Co., Ltd., India) against PBS to form micelles. The copolymer concentration in the micelle preparations was 0.5 % (m/v).

To obtain nanodroplets, 2 % PFP (v/v) was introduced into the micellar solution, which was then sonicated for 90 s by a 20-kHz ultrasonic processor (Ultrasonic Homogenizer, Cole Parmer, UK). Sonication was performed in an ice bath to minimize the vaporization of PFP.

2.3. Characterization of nanodroplets

The particle size distribution and zeta potential of the nanodroplets were measured by a Zetasizer Nano ZSE instrument (Malvern Instruments Ltd., Malvern, UK). The nanodroplets were heated at 37 °C for 1 h, during which the particle size distribution was measured every 10 min. In addition, the size distribution of nanodroplets stored at 4 °C

was measured. The mean value of three independent measurements was designated as the mean particle diameter.

Transmission electron microscopy (TEM, HT7700, HITACHI, Japan) was used to observe the morphology of the nanodroplets. Before observation, nanodroplet dispersions were deposited onto copper grids and then stained with uranyl acetate.

2.4. Phase transition and contrast-enhanced imaging of nanodroplets *in vitro*

PFP droplets (0.5 % PEG-PCL, 2 % PFP) were prepared using the method described in Section 2.2 and diluted 4 times with degassed distilled water. Plastic pipettes purchased from Kangwei Medical Devices Co., Ltd. (Taizhou, China) were filled with the diluted droplets solution and sealed by clamps. Pipettes filled with degassed distilled water was used as control. The size of the pipette used for imaging is 3 cm in length, 8 mm in diameter and 1.5 ml in volume. A large beaker paved with a layer of sound-absorbing sponge at the bottom was filled with degassed distilled water and heated with a water bath. The temperature of the water in the beaker was monitored with a thermometer. After reaching the desired temperature, the pipette with samples was immersed into the beaker, and ultrasound imaging was conducted 10 min later. The selected temperatures were 25 °C and 37 °C.

A Hivision Ascendus ultrasound diagnostic system (HITACHI, Japan) equipped with a L74 M linear array transducer (5–13 MHz) was used to conduct *in vitro* imaging experiments. The frequency used was 13 MHz, and the imaging acquisition type under contrast mode was pulse inversion. The ultrasound transducer head coated with coupling gel and preservative film was maintained parallel to the long axis of the pipette in water. A single focus was used, and the focus was placed between 1.25 cm and 1.5 cm. The imaging target depth was 4 cm. The ultrasound contrast-enhanced imaging was observed in real time. The MI ranged from 0.08 to 1.0, and the imaging was repeated three times for each MI. In addition, *in vitro* contrast-enhanced imaging of the nanodroplets stored at 4 °C for 2 weeks was conducted using the same method.

2.5. Contrast-enhanced imaging of nanodroplets *in vivo*

2.5.1. Animal procedures

Four- to six-week-old male Kunming mice with body weights of 22–24 g were purchased from SPF Biotechnology Co., Ltd. (Beijing, China) and maintained in accordance with the National Institutes of Health Guidelines for the Care and Use of Laboratory Animals. All animal experimental procedures were approved by the Animal Ethics Committee of Henan University. The animals were housed in cages (320 × 215 × 170 mm) before experiment, 5–8 mice per cage. The temperature and relative humidity of the animal room were set to 23 ± 3 °C and 55 ± 15 %, respectively. The ventilation frequency was 10–20 times per h. The mice were divided into four groups, five in each group. The mice were anaesthetized by an intraperitoneal injection of 10 % (m/v) chloral hydrate (4 ml/kg), which could provide 2 h of anaesthesia. Then, the mice were placed on their backs on heated pads with their limbs fixed, and the hair on the chest was removed. Contrast-enhanced imaging of mice livers were performed before injection of contrast agents. Then, one group was intravenously injected with 100 µl SonoVue®, and the MI selected was 0.08 in the contrast-enhanced imaging experiment. The other three groups were intravenously injected with 100 µl nanodroplets, and the MI used in contrast-enhanced imaging was 0.08, 0.6 and 1.0.

2.5.2. Contrast-enhanced imaging of mice livers

An Arietta 70 ultrasound diagnostic system (HITACHI, Japan) equipped with a L441 linear array transducer (2–12 MHz) was used to conduct the *in vivo* imaging experiments. The frequency used was 10 MHz, and the imaging acquisition type under contrast mode was

amplitude modulation. A single focus was used, and the focus was placed between 0.75 cm and 1.0 cm. The imaging target depth was 2 cm. These parameters were consistent throughout the course of the experiment. Contrast-enhanced imaging of mice livers were observed and recorded at the same time as the injection of nanodroplets or SonoVue, which was regarded as the starting point for imaging of the contrast agents. The time duration of recording was 2 min for the initial period. After injection, the contrast-enhanced imaging of mice livers were observed and recorded every 10 min, and the time duration of recording was 1 min. The last time point for observation was 1 h after injection.

2.6. Intensity analysis of contrast enhancement for nanodroplets

The contrast enhancement intensity of the nanodroplets in the mice livers were analysed using DAS software (HITACHI, Japan) installed on the Arietta 70 ultrasound diagnostic system. The recorded ultrasonic dynamic imaging was loaded into the software, and circular regions of diameter 3 mm were selected in the part of the liver parenchyma with uniform vascular distribution, namely, the regions of interest (ROIs). Then, the acoustic signal intensity within the ROI over time was analysed by DAS software, and a curve of the intensity over time was obtained. The liver area where the ROI was located was consistent for all mice in the intensity analysis.

2.7. Statistical analysis

The results are presented as the mean $\pm$ standard deviation (SD) for at least three independent samples; the S are indicated as error bars in each graph. Statistical analyses of the experimental data from different groups were performed by one-way ANOVA. A value of $p < 0.05$ was considered significant, and $p < 0.01$ was considered highly significant.

3. Results and discussion

3.1. Particle size distribution and nanodroplet stability

The particle size distributions of the micelles and nanodroplets at 25 °C were presented in Fig. 1. The average diameters of the PEG-PCL micelles and nanodroplets were approximately 41.2 ± 1.1 nm (Fig. 1A) and 391.0 ± 6.3 nm (Fig. 1B), respectively. The zeta potential of the nanodroplets was -7.18 ± 0.42 mV. It has been reported that the composition of the nanodroplets solution depends on the copolymer/PFP ratio, which can be expressed as one of three situations [18]: (1) micelles only (no nanodroplets), PFP dissolved in the micelle cores; (2) micelles coexisting with the nano- or microdroplets; and (3) nanodroplets only (no micelles). In Fig. 1B, most of the particles were

distributed in the range of 100–1000 nm, and no particles smaller than 100 nm were found, which indicated that there were only nanodroplets in the solution.

The stability of the nanodroplets at 37 °C was investigated. The results illustrated in Fig. 2A implied that the nanodroplets showed two distribution peaks after heating at 37 °C for 1 h. The median particle sizes corresponding to each peak were 121.3 ± 24.8 nm (peak 1) and 777.6 ± 16.7 nm (peak 2). The average particle size increased gradually from 391.0 ± 6.3 nm to 472.9 ± 9.1 nm, approximately 20 %. The particle size increase of the nanodroplets at 37 °C could be attributed to nanodroplet/nanobubble conversion. The boiling point of PFP is 29 °C, so nanodroplets were converted into nanobubbles or microbubbles due to the vaporization of PFP upon heating to physiological temperature [18]. These nanodroplets/nanobubbles were mainly represented in peak 2. However, not all of the nanodroplets underwent phase transition. The surface tension provided by the droplet shells form a pressure difference between the internal and external environment of the droplet, that is, the Laplace pressure [20–22].

$$\Delta P_{\text{Laplace}} = P_{\text{Inside}} - P_{\text{Outside}} = 2\sigma/r \quad (1)$$

where P_{Inside} is the pressure inside the droplet, P_{Outside} is the ambient pressure in the surrounding media, r is the radius of the droplet, and σ is the surface tension.

The Laplace pressure is an inverse function of the radius, which suggests that decreasing the droplet diameter will increase the vaporization threshold [23]. Therefore, the PFP inside smaller nanodroplets could remain in the liquid state even at physiological temperature. The particles that comprised peak 1 in Fig. 2A were mainly smaller nanodroplets with high vaporization threshold.

Though phase transition occurred in a portion of the nanodroplets at physiological temperature, the slight change in the average particle size of the nanodroplets would not affect accumulation in the tumour. It has been reported that nanocarriers with particle size between 380 nm and 780 nm can be passively targeted to tumour sites through large inter-endothelial gaps [24,25], and most of the particles in Fig. 2B were within this range.

The particle size distribution of the nanodroplets stored at 4 °C for 50 h was measured. As shown in Fig. 3, there were three peaks in the diagram, and the corresponding particle sizes of the peaks were 41.3 nm, 200.2 nm and 2321 nm. The peak centred at 41.3 nm corresponded to the polymer micelles, and the remaining two peaks corresponded to droplets. The results implied that the nanodroplets became more polydispersed during the preservation process due to the reduction in stability. Small droplets coalesced to form large droplets, increasing the number of large droplets. In addition, the PFP in some of the droplets escaped, leaving only the micelles.

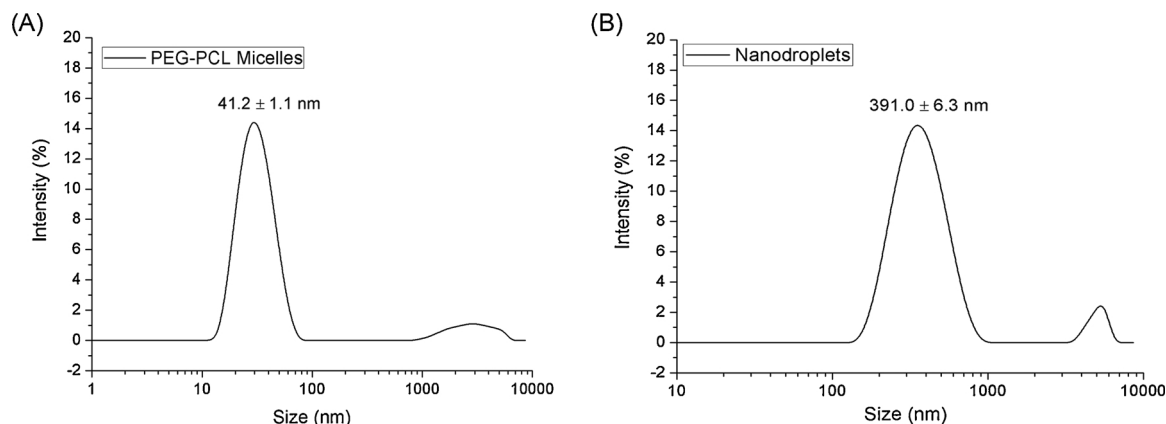


Fig. 1. The particle size distributions of PEG-PCL micelles (A) and nanodroplets (B) at 25 °C ($n = 3$, mean $\pm$ SD).

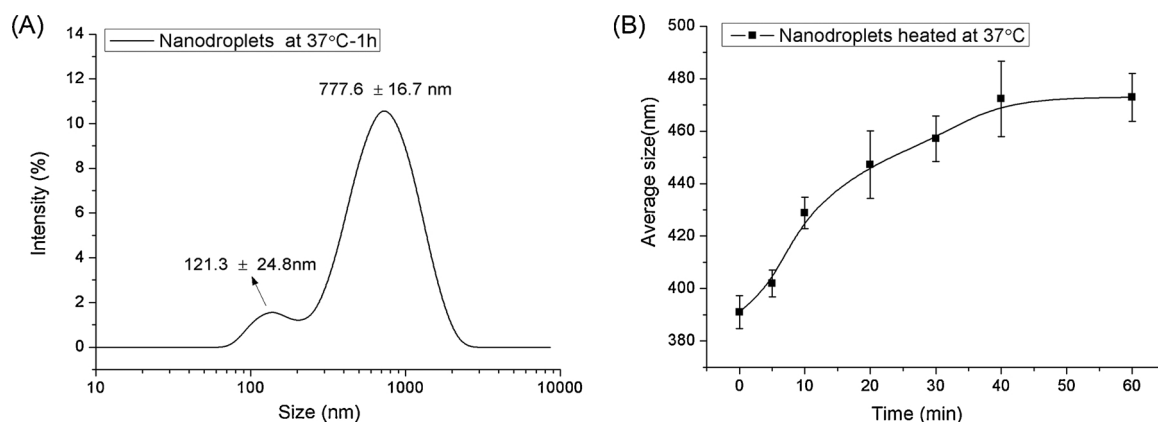


Fig. 2. The particle size distribution of nanodroplets heated at 37 °C for 1 h (A) and variation of average particle size with time (B) ($n = 3$, mean $\pm$ SD).

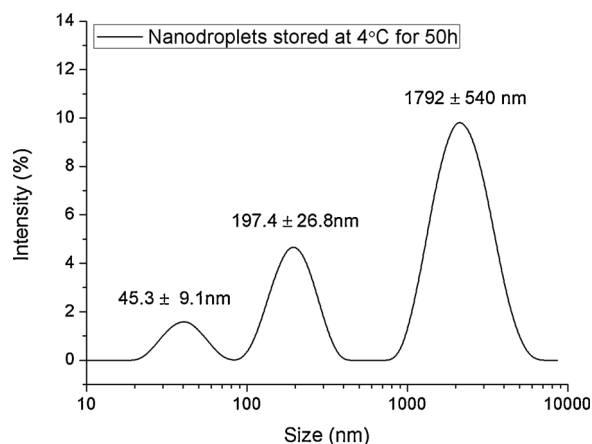


Fig. 3. The particle size distribution of nanodroplets stored at 4 °C for 50 h ($n = 3$, mean $\pm$ SD).

3.2. Morphological characterization of nanodroplets

Fig. 4A and B showed two representative morphology of parallel nanodroplets samples with same dilution observed by TEM. The particle diameters of both were smaller than 1 μm . In Fig. 4A, the copolymer molecules are used for droplet stabilization, and the micelles are destroyed. Upon introduction of PFP into the micellar solution and sonication, some copolymer molecules were transferred from the micelles onto the droplet surface to form droplet walls, which resulted in

gradual micelle degradation [18]. The nanodroplets in Fig. 4B showed a clear shell-core structure. PFP droplets were introduced into the hydrophobic core of the PEG-PCL micelles, and the micellar structure was maintained.

3.3. Phase transition and contrast-enhanced imaging of nanodroplets *in vitro*

Different temperatures and MIs were selected to investigate the contrast-enhanced imaging of nanodroplets *in vitro*, and the results were shown in Fig. 5. At 25 °C, no acoustic enhancement was observed when the MI varied from 0.08 to 1.0, indicating that the phase transition did not occur at temperatures below the boiling point of PFP. At 37 °C, acoustic signals could be observed when the MI reached 0.4, and the acoustic enhancement was stronger for higher MIs. In addition, the formation and movement of bubbles was observed. At MI of 0.4, bubbles mainly appeared from the bottom of the pipette and gradually moved upward. At MI ≥ 0.6 , bubbles appeared both from the bottom and the top of the pipette. Then bubbles at the bottom moved upward, and bubbles at the top was pushed downwards, which should be caused by the radiation force of ultrasound. Finally, nearly all of the bubbles stayed on the inner wall of the pipette. No obvious sign of bubble destruction was observed. These results suggested that ultrasound stimulation played an essential role in the liquid-to-gas transition of the nanodroplets.

The boiling point of PFP at atmospheric pressure is 29 °C. However, when the PFP is stabilized by block copolymer walls, the droplets experience an additional pressure that prevents vaporization of the nanodroplets due to interfacial surface tension effects, defined as the

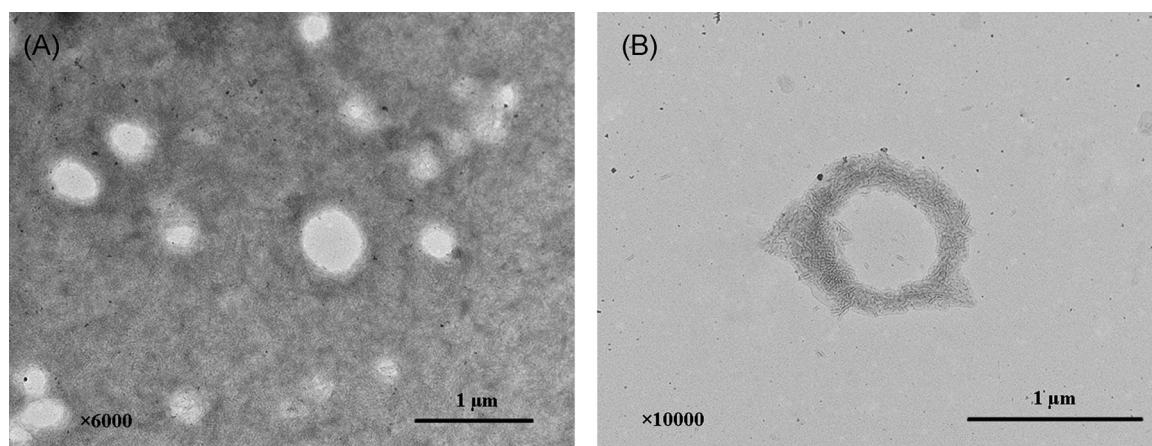


Fig. 4. TEM images of nanodroplets: (A) nanodroplets embedded in polymers; (B) nanodroplets with shell-core structure. These are representative images for three independent samples.

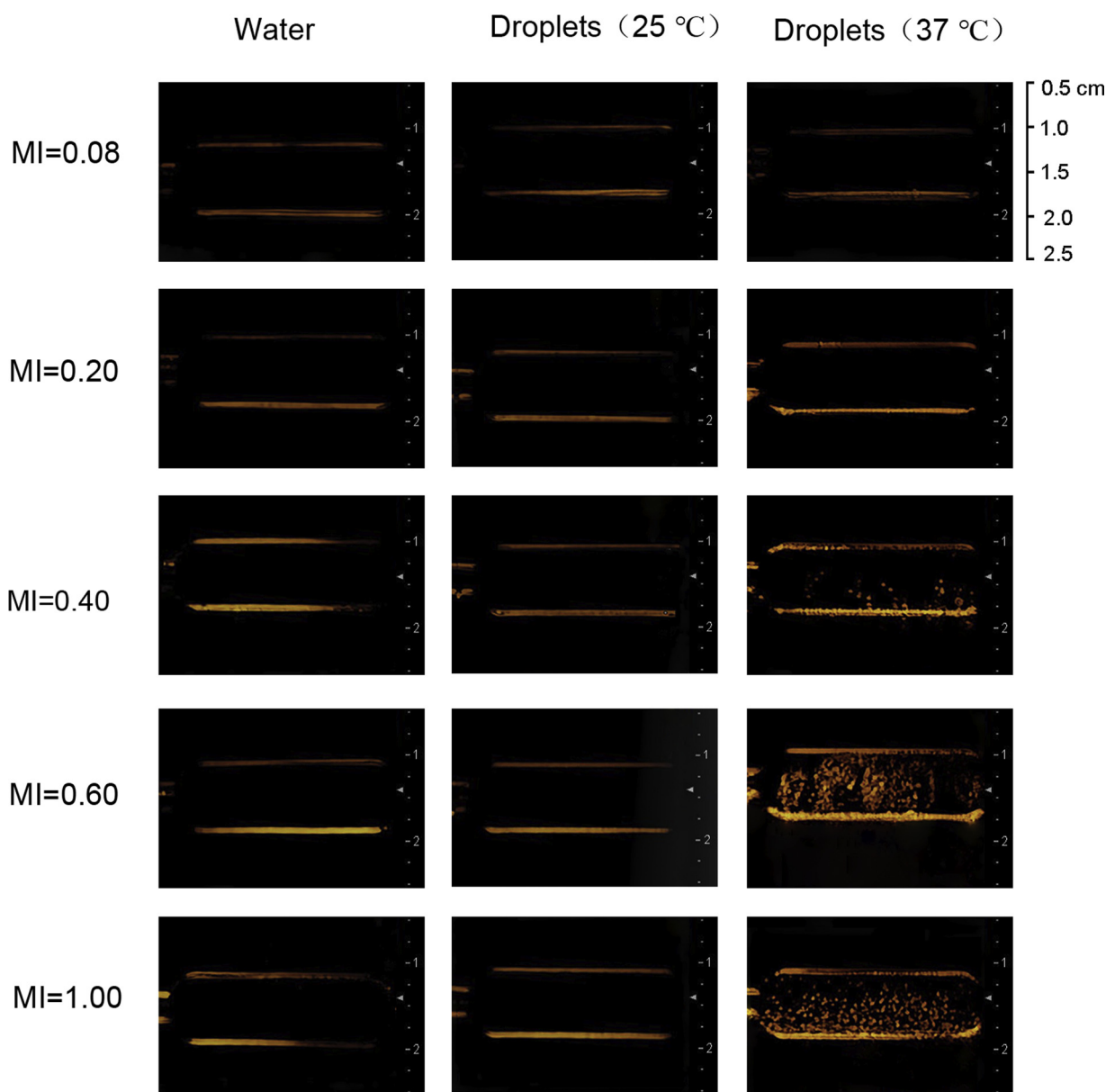


Fig. 5. Contrast-enhanced ultrasound imaging of nanodroplets at 25 °C and 37 °C *in vitro* with MI varying from 0.08 to 1.0.

Laplace pressure. The Laplace pressure is an inverse function of radius, and the ultrasonic energy required to induce phase transition increases when the pressure is raised at the droplet core [23], which suggests that increasing droplet diameter will decrease the vaporization threshold. At low MI ($MI = 0.4$), the ADV or thermal droplet vaporization of droplets with large particle size generated nanobubbles or microbubbles, and weak enhancement was produced. When a higher MI was applied, the ultrasonic energy was sufficient to activate the ADV of more nanodroplets with smaller size, and the acoustic enhancement became stronger.

It has been reported that the pressure amplitudes required to induce the phase transition of acoustically sensitive droplets seem to be substantial, with MI as high as four, well above the FDA-approved limit for diagnosed ultrasound [12]. Related studies tend to use focused ultrasound to activate the vaporization of the nanodroplets [17]. However, the nanodroplets prepared in this study could be activated to vaporize by the diagnostic ultrasound commonly used in clinic, and the MI was within the normal range. This may be due to the relatively thin and soft droplet shell made of micelles, reducing the pressure amplitudes required. In addition, the *in vitro* imaging performance of the

nanodroplets stored at 4 °C for two weeks was almost unaffected.

The exact mechanism that induces the phase transition of the droplets as a result of ultrasound energy has been discussed in related studies, and a number of mathematical and physical models have been developed to simulate droplets vaporization [26–28]. The ultrasound pressure required to induce the phase transition of the droplets is affected by a large number of factors, which can be divided into three categories: environmental factors, droplet design, and ultrasound parameters [23]. One hypothesis is that the phase transition occurs due to acoustic heating of the droplet, but most studies have shown that increasing the ultrasonic pulse duration resulted in no significant decrease in the required ultrasound, which indicated that heating may not be the primary mechanism at work [29]. Therefore, nanodroplets heated at 37 °C could still remain stable under ultrasonic irradiation with low MI. Other studies have provided evidence that ADV may be initiated by mechanical effects such as acoustic and hydrodynamic cavitation [23]. These two kinds of cavitation are caused by bubbles in water under the change of acoustic pressure and water pressure respectively. These cavitation could produce similar mechanical effects such as micro-streaming, liquid jet and shock wave [5], which may

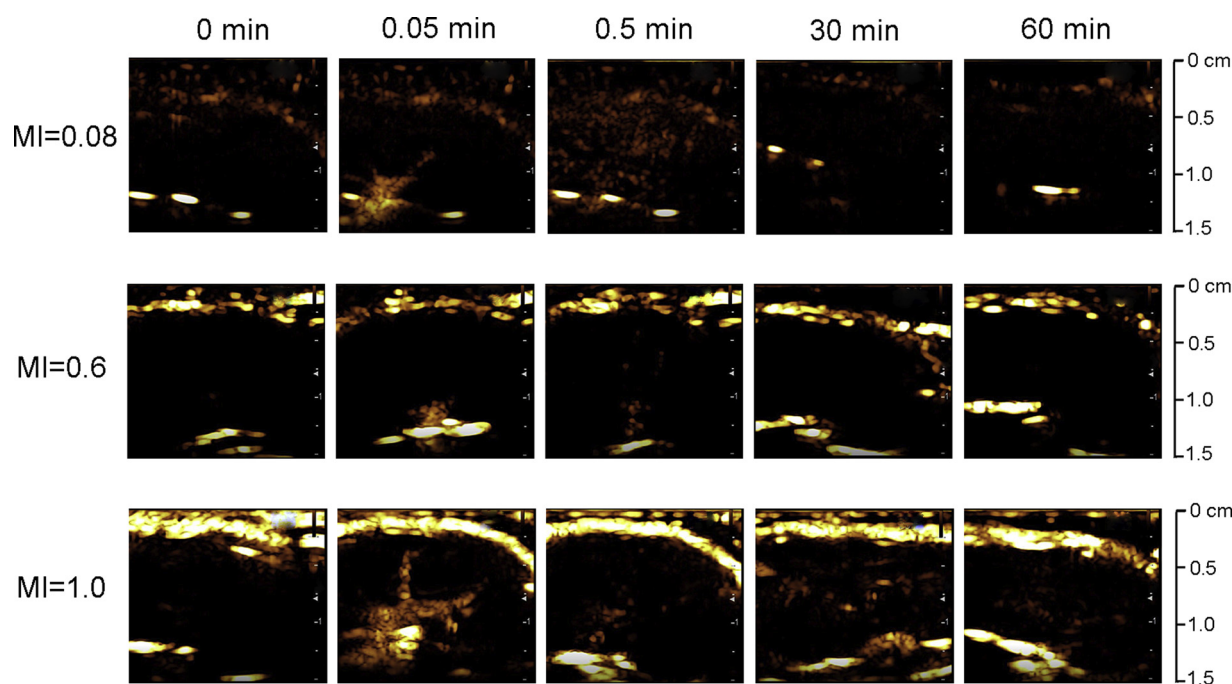


Fig. 6. Contrast-enhanced ultrasound imaging of nanodroplets in mice livers at MI of 0.08, 0.6 and 1.0 within 1 h.

impinge on the surface of droplets and induce vaporization [30]. However, more studies have shown that vaporization can be induced independently of inertial acoustic cavitation [31,32]. A recent interpretation validated with experimental data captured with an ultrahigh-speed camera states that ADV was initiated by a combination of two phenomena: highly nonlinear distortion of the acoustic wave before it hits the droplet and focusing of the distorted wave by the droplet itself [33]. At high excitation pressures, nonlinear distortion causes significant superharmonics with wavelengths on the order of the droplet size. The superharmonic focusing of acoustic energy inside the droplet causes a spot of negative pressure that spreads throughout the liquid volume [34]. This could explain the phase transition of nanodroplets under high MI.

3.4. Contrast-enhanced imaging of nanodroplets in mice livers

MI of 0.08, 0.60 and 1.0 were selected to conduct contrast-enhanced imaging of nanodroplets in mice livers (Fig. 6). The imaging of nanodroplets *in vivo* was quite complicated and different from that *in vitro*. At the initial stage after injection (0.05 min), contrast enhancement could be seen in the portal vein for all MI, which indicated that nanobubbles/microbubbles had been formed. At 0.5 min, the acoustic signals almost completely filled the whole livers at the MI of 0.08, but only a small amount of signals can be seen at MIs of 0.6 and 0.8. After 20 min, the signals disappeared for MIs of 0.08 and 0.6. While the signals for the MI of 1.0 lasted more than 1 h, and the intensity was stronger than that of the initial stage.

The contrast enhancement performance of the nanodroplets in the mice livers was studied by analysing the acoustic signal intensity within the ROI located in liver parenchyma with uniform vascular distribution. The results in Fig. 7 suggested that the intensity of contrast enhancement increased significantly for all MI at the initial stage (2 min) after injection of the nanodroplets (Fig. 7A), and the intensity for the MI of 0.08 was strongest among all the MIs (Fig. 7B). At 30 min and 60 min, the intensity was reduced to the same level as before injection for MIs of 0.08 and 0.6. However, the acoustic signals lasted for more than 1 h for the MI of 1.0, and the intensity at 30 min and 60 min was stronger compared with that at the MIs of 0.08 and 0.6. Moreover, the intensity for the MI of 1.0 at 60 min was stronger than that at 2 min.

The acoustic signals at the MI of 0.08 may result from the phase transition of droplets with larger particle size, which had a lower Laplace pressure and vaporized under low pressure amplitudes or physiological temperature. This also explained the acoustic signals in the portal vein for all MIs at the initial stage, but the weaker signal intensity of the MIs of 0.6 and 1.0 may result from the rapid destruction of larger microbubbles under high pressure amplitudes.

The continuous imaging at the MI of 1.0 was due to the nanoscaled droplets, which needed higher acoustic pressure to vaporize and had a long circulation time. Bubbles resulting from droplets with smaller size will experience a greater Laplace pressure, which will serve to compress the gas core to a greater extent, so less echogenic and limited ultrasound contrast was provided [35]. Studies on polymer-coated PFP nanodroplets have suggested that a significant amount of nanodroplets were still in circulation as much as 4–5 h after intravenous injection in mice [36,37]. Moreover, these nanodroplets easily accumulated in reticuloendothelial organs, such as the liver and spleen, which were then recognized and phagocytosed by macrophages or simply trapped within these organs [17]. If the acoustic pressure was enough to induce a phase transition, the enhanced acoustic signals could last for a long time. For MIs of 0.08 and 0.6, the smaller droplets could not vaporize, so there were nearly no signals after 10 min.

The obvious acoustic signals generated from vaporized droplets at a very low MI (0.08) were noteworthy, so the imaging performance of the nanodroplets was further compared with that of SonoVue® (Fig. 8A). For these two contrast agents, the acoustic signals filled the whole livers immediately after injection and were almost cleared within 10 min. The signal intensity curves of these two contrast agents, shown in Fig. 8B, were consistent in trend and peak time (20 s), except for the stronger intensity from SonoVue®. By the ideal gas law with surface tension included, once a PFC droplet is vaporized, the resulting bubbles will theoretically expand approximately 3–6 times in diameter [21,28]. According to the particle size of the nanodroplets, the bubbles generated from the phase transition were composed of nanobubbles and small microbubbles. However, SonoVue® has a core of sulfur hexafluoride, and 90 % of the particles had a diameter smaller than 6 μ m. The smaller particle size of the vaporized bubbles may be one reason accounting for the weaker signal intensity. Besides, the low concentration may also be the reason. According to the study of Gao et al.

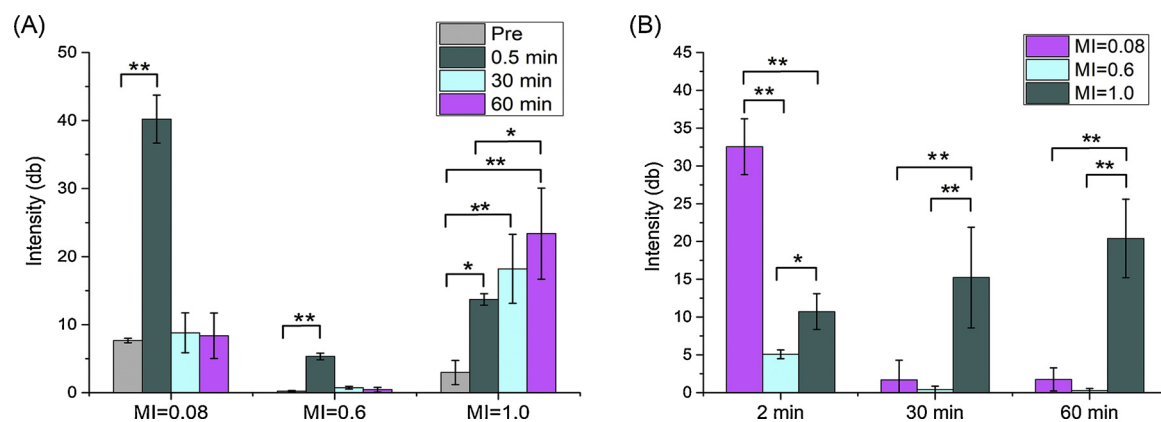


Fig. 7. Contrast enhancement intensity of nanodroplets in mice livers for MI of 0.08, 0.6 and 1.0. (A) Contrast enhancement intensity within ROI before injection and at different time after injection for each MI. (B) Contrast enhancement intensity within ROI for different MI at same time point after injection of nanodroplets (the intensity before injection was deducted). (n = 3, mean $\pm$ SD. * P < 0.05, ** P < 0.01).

[18], nanodroplets prepared using the same materials and methods as the presented study have concentration of 10^8 – 10^9 /ml. However, only a portion of the nanodroplets vaporized at MI of 0.08. Therefore, the actual concentration of microbubbles vaporized from droplets may be lower than SonoVue® ($2-5 \times 10^8$ /ml) at the same MI.

The health status of the mice was monitored during the

experimental process, and no abnormal body temperature, breathing difficulties or sudden death was observed, which confirmed the safety of the nanodroplets as ultrasound contrast agents.

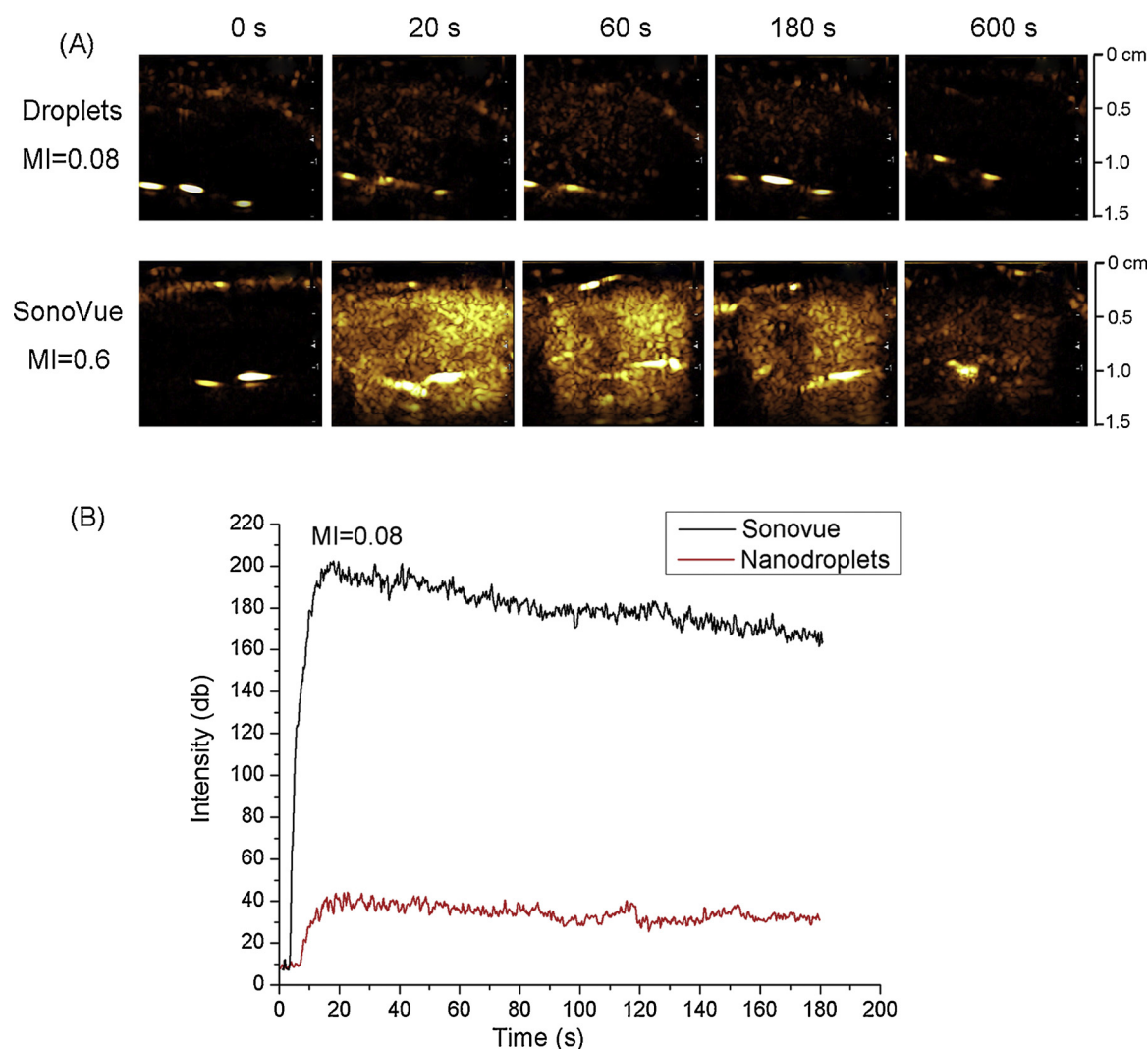


Fig. 8. Contrast-enhanced imaging of nanodroplets and SonoVue® in mice livers at MI of 0.08 (A) and intensity of contrast enhancement over time (B).

4. Conclusions

In this study, the phase transition and *in vitro/in vivo* imaging of polymeric nanodroplets with PFP core were studied in depth, and the phase transition mechanism of the nanodroplets was elucidated. The size measurements showed that the average particle diameter of the nanodroplets increased and the size distribution became more dispersed after heating at 37 °C due to the phase transition of some nanodroplets with lower vaporization thresholds. The results of the *in vitro* imaging showed that temperature and MI both affected the phase transition of the nanodroplets. Below the boiling point, a high MI could not induce vaporization of the droplets. Above the boiling point, the nanodroplets vaporized when the MI was greater than 0.4. The analysis of contrast enhancement intensity showed that the signal was strongest for the low MI (0.08) in the initial stage, but only acoustic signals at the MI of 1.0 lasted for a long time. Because droplets with smaller size have longer circulation time and can only be induced to vaporization by high acoustic pressure. The above study on the phase transition mechanism and contrast-enhanced imaging could provide a basis for the development and application of ultrasound-responsive nanodroplets in clinic.

CRediT authorship contribution statement

Yaqiong Li: Conceptualization, Methodology, Investigation, Visualization, Writing - original draft. **Ruiqing Liu:** Investigation, Validation, Data curation. **Luwen Liu:** Investigation, Software. **Ye Zhang:** Software, Formal analysis. **Jun Sun:** Writing - review & editing. **Peizhi Ma:** Project administration. **Yuejin Wu:** Resources. **Shaobo Duan:** Software, Formal analysis. **Lianzhong Zhang:** Conceptualization, Supervision, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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